Music-Gen v3 SPINE Milestone — Fanout Clone 2: WIG Palette Render (Cycles 1–2)

2026-09-03

Contents

Music-Gen v3 SPINE Milestone — Fanout Clone 2: WIG Palette Render (Cycles 1–2)

Abstract

This report covers Cycles 1 and 2 of a fanout-clone branch that extends the c21 Chicken Grease palette-render proof (`PALETTE_MOVES_PANEL`, verdict SHA `5ba4eaca242fcd29…`) to What If I Go (source SHA-16 `252eb21ce7df7328`) as the second focus-song palette A/B, satisfying the operator D-D 2026-09-02 pre-condition (two-song palette proof) that gates the palette-becomes-primary campaign-wide flip. The clone (fork `4c826786aced`, clone 2) was assigned to run the same palette pipeline as c21 verbatim (drums fluidsynth channel 10; bass Surge XT VST3 via DawDreamer with c33 P1 iterate-params hydration and REDEFINED_GAP fallback on c36-envelope breach; guitar/piano/other sfizz with `fluidsynth_gm` fallback on `sfz_dir_missing`; vocals verbatim D2 htdemucs copy from c21 WIG delivery), apply the c6 Method B iirpeak-EQ+RMS+LUFS-S loudness chain via READ-ONLY import, and deliver a sibling reconstruction under `data/v3/deliveries/252eb21ce7df7328/palette_render_c25/`. Cycle 1 landed the rubric doc `docs/v3_spine_wig_palette_render_c25_rubric.md` (SHA `80fc4b60bbc475738b8dc641e9d698f4f0c1bacb923b8b2eca40c9a8c01a1a50`) mtime-hard before any script under `scripts/v3_spine/palette_render_wig/`, executed the pipeline against the WIG operator section t = 72.77133786848073 s – 102.77133786848073 s from `focus_set_v2.json`, and emitted verdict SHA `e8285ceed4c133b618a1085040d663096c5506a33665744d5ba121039f17511b` with `PALETTE_MOVES_PANEL` firing on **5-of-5** numeric panel keys exceeding the 5% relative-delta Comparison B threshold (all five: mel_L1 48.66%, spectral centroid 42.99%, RMS-env 89.64%, LUFS-M 86.79%, VGGish 34.42%). Cycle 2's auditor performed live SHA re-verification on every load-bearing anchor with byte-exact match and closed the branch with `COMPLETE` and `[[BRANCH_COMPLETE]]` under the `<no-null-cycle-validation>` rule. The load-bearing honesty caveat that governed the c21 Chicken Grease palette proof applies verbatim to this branch: all six stems ended up on fluidsynth GM at the bottom of their respective fetchability ladders

(bass Surge XT hit `max_pairwise_rms = 0.041` — 410× outside the c36 clone-2 envelope of 1e-4, engaging the REDEFINED_GAP arm to fluidsynth GM(33); guitar/piano/other sfizz `sfz_dir_missing` → fluidsynth GM(25/0/88)), so the measured palette movement is at the mechanism level fluidsynth GM plus program substitution plus Cycle 6 Method B 12-band iirpeak EQ plus per-stem RMS/LUFS-S loudness match, not genuine Surge XT/sfizz timbral character. Operator D-D pre-condition is now satisfied on both Chicken Grease and What If I Go; the palette-becomes-primary flip remains gated on operator ear confirmation of audible improvement on either A/B pair.

1. Introduction and scope

The operator's D-D directive of 2026-09-02 named a two-song palette-render proof as the pre-condition for a campaign-wide palette-becomes-primary flip: if a Surge XT VST3 + sfizz palette render moves the perceptual panel measurably on at least two operator-approved focus songs and the operator confirms audibly-different-AND-audibly-better on the A/B, palette becomes primary and all remaining focus songs re-render under the palette pipeline as secondary deliverables. Cycle 21 clone-2 landed the first proof on Chicken Grease at verdict SHA `5ba4eaca…5644a` with `PALETTE_MOVES_PANEL` firing on 4-of-5 keys. This branch is the second proof on What If I Go — the "second focus-song A/B pair" the operator now has ear-material for.

Sibling clones in the same fork (`4c826786aced`) run parallel opening work: clone 0 executes the c25 Peach Dream detached-launch checkpointed driver resume; clone 1 executes the c25 one-off driver retirement contract per the c22 catalog. Per the audit's cumulative notes, clone 1 carries a CRITICAL fabricated `reproduce_proof_authorization` SHA-citation bookkeeping defect (underlying data OK; citation broken); it is audited on its own merge path and is out of scope here.

The clone's scoped objective as issued:

- **Pre-register the rubric** `docs/v3_spine_wig_palette_render_c25_rubric.md` mtime-hard before any script under `scripts/v3_spine/palette_render_wig/`, with a three-way `rubric_hash_v2` chain (doc SHA == `data/v3_spine/252eb21ce7df7328/palette_render/rubric_hash_v2.txt` content == verdict field).
- **Consume the operator section** for WIG from `focus_set_v2.json`.
- **Per-stem dispatch mirrors c21 verbatim** via READ-ONLY imports of `scripts/v3_spine/palette_render/*` plus `scripts/v3_spine/rc7_v2_rerun_v3_paths.py`: drums fluidsynth channel 10; bass Surge XT VST3 with c33 P1 iterate-params hydration and REDEFINED_GAP fallback on c36-envelope breach (c31 STILL_GAP + c35 A anti-patterns remain locked; no re-attempt of VST3 state APIs); guitar/piano/other sfizz with `fluidsynth_gm(25/0/88)` fallback on `sfz_dir_missing`; vocals verbatim D2 htdemucs copy from c21 WIG delivery.
- **Apply the c6 Method B iirpeak EQ + RMS + LUFS-S loudness match chain** verbatim.
- **Byte-determinism ×2 gate** per persisted stem WAV under env pins.
- **Measure both panels 8-key finite**: Comparison A (original vs palette-render) and Comparison B (c21 WIG operator-blessed fluidsynth render vs palette).
- **Verdict enum**: `PALETTE_MOVES_PANEL` fires when Comparison B delta magnitudes exceed 5% relative threshold on ≥3/5 numeric keys; `PALETTE_NEUTRAL` when <3/5; `RENDER_FAILS` on any byte-det ×2 halt without documented REDEFINED_GAP arm.
- **Sibling delivery** at `data/v3/deliveries/252eb21ce7df7328/palette_render_c25/`.
- **c21 WIG operator-blessed delivery byte-identical pre==post** (anchor snapshot ≥30 SHAs including manifest `9a8a09d0f553a79f…` + Chicken Grease c21 palette anchors + all preserved v3-spine scripts).

- **Six named + two housekeeping ledger events** under the `-clone-2` suffix on infra families; substantive `M-V3-SPINE-1/wig-palette-render-c25` row unsuffixed per c32.
- **Test suite ≥14 cases** covering the fourteen named invariants (rubric mtime pre-reg, three-way rubric chain, render_stem SHA lock, no-PRNG grep, VST3 state API AST forbidden, `/usr/bin/python3` guard, c48 env-flag defaults OFF, focus_set_v2 consumption for WIG, both panels 8-key finite, cross-song anchor preservation, byte-determinism ×2 per stem, honest REDEFINED_GAP arm bookkeeping, dispatch summary matches fetchability ladder, delivery manifest carries env_pins block).

The required output artifact is `docs/v3_focus_wig_palette_render_c25_report.md`.

2. Cycle 1: WIG palette-render pipeline execution

2.1 Rubric freeze (before any script)

`docs/v3_spine_wig_palette_render_c25_rubric.md` (SHA `80fc4b60bbc475738b8dc641e9d698f4f0c1bacb923b8b2eca40c9a8c0` was committed mtime-hard before any script under `scripts/v3_spine/palette_render_wig/`. Its pinned hash file at `data/v3_spine/252eb21ce7df7328/palette_render/rubric_hash_v2.txt` (65 bytes, trailing newline included) carries the same SHA verbatim. The rubric defines the three-verdict enum, the Comparison B threshold (≥3-of-5 numeric keys exceeding 5% relative delta versus the c21 WIG-vs-original reference panel), the byte-determinism ×2 mandatory sub-clause with the c36 REDEFINED_GAP fallback envelope for Surge XT bass, and the sfizz-fallback ladder for the SFZ-driven stems.

2.2 Fetchability ladder and per-stem routing

The pipeline executed the fetchability ladder for each melodic stem, honestly recording every routing decision at `data/v3/deliveries/252eb21ce7df7328/palette_render_c25/fetchability_ladder.jsonl`. Outcomes summarized in the verdict as `sfizz_fallback_reason: sfz_dir_missing_no_sfz_files_in_workspace` and `sfizz_fallback_stems: [guitar, piano, other]`:

Stem	Intended path	Actual path	Fallback reason
bass	Surge XT VST3 via DawDreamer (c33 P1 iterate-params)	fluidsynth GM(33)	Byte-determinism ×2 failed with `max_pairwise_rms = 0.041` (410× outside c36 clone-2 envelope of 1e-4). REDEFINED_GAP arm engaged. c31 STILL_GAP + c35 A anti-patterns preserved — no re-attempt of VST3 state APIs.
guitar	sfizz via `sfizz_render` CLI	fluidsynth GM(25)	`sfz_dir_missing_no_sfz_files_in_workspace`
piano	sfizz via `sfizz_render` CLI	fluidsynth GM(0)	`sfz_dir_missing_no_sfz_files_in_workspace`

Stem	Intended path	Actual path	Fallback reason
other	sfizz via `sfizz_render` CLI	fluidsynth GM(88)	`sfz_dir_missing_no_sfz_files_in_workspace`
drums	fluidsynth GM channel 10 (c21 pattern)	fluidsynth GM channel 10	(intended)
vocals	htdemucs D2 verbatim (from c21 WIG)	htdemucs D2 verbatim	(intended)

Every stem ended up on fluidsynth GM at the bottom of its fetchability ladder — the same shape as the c21 Chicken Grease palette proof, and for the same two reasons: Surge XT VST3 binary carries an internal nondeterminism envelope larger than the campaign's acceptance threshold on this specific input path (0.041 versus 1e-4), and the sfizz sample-library path is unpopulated in the current workspace. This is a first-class honest disclosure per Fixed Decision 1, not a failure to smooth over.

2.3 Per-stem palette renders and c6 Method B loudness chain

Per-stem WAVs landed under `data/v3/deliveries/252eb21ce7df7328/palette_render_c25/per_stem/`. Each was rendered twice into fresh temporary directories under identical environment pins with the SHA-256 asserted equal across the two runs. Canonical MIDI SHAs for the seven probes serialized via c4 read-only:

Probe	Canonical MIDI SHA
bass	`5562e3630dfce06460db04f3d9a5c0f552441a65c39c71010b49386b906f4`
drums	`7403f8f383da5499116186dfd52084f1927de8b562c33b2c8f1e933bd662f`
guitar	`0c171b00b141daef90e010c52676304d75c558f93693326c7553caef1bb95`
other	`a7ccbf5755f43fe73d591fd919604e5a0ab769bd94097313d47765c7285da`
full_mix	`0d15c3c66fd2a6776ca649ff1b900d20b8039c202b78b4b58735a47189eb0`
piano, vocals	(per-probe SHAs pinned in verdict `sub_artifact_shas.canonical_midi`)

The Cycle 6 Method B rc7 12-band iirpeak EQ + RMS + LUFS-S loudness match was applied verbatim through a READ-ONLY import of `scripts/v3_spine/rc7_v2_rerun_v3_paths.py` (SHA `eaaa993e2eb50d25a9085af0b1171bc58da9a9c21b6233cc9c0c80b1c6f03e38`, unchanged). The palette-rendered full reconstruction landed at `data/v3/deliveries/252eb21ce7df7328/palette_render_c25/full_reconstruction_palette.wav` (SHA `fd47390ae41a58867f6bf1fd493dac61e18290feaedcb134bf715ede43fcc0ea`), sibling to the c21 WIG operator-blessed `data/v3/deliveries/252eb21ce7df7328/operator_section/` delivery. The c21 delivery is preserved byte-identical.

2.4 Panel measurement

The 8-key perceptual panel was measured on both required comparisons with all five reported numeric keys finite on both.

Comparison A (original vs palette-render) at `panel_original_vs_palette.tsv`:

Key	Value
spectral_centroid_rmse_hz	1 090.92
mel_l1_db	10.228
rms_env_rmse	0.18871
lufs_m_rmse_lu	9.251
embedding_cosine_distance (VGGish)	0.08477

Comparison B (c21 WIG fluidsynth vs palette-render) at `panel_fluidsynth_vs_palette.tsv`:

Key	Value
spectral_centroid_rmse_hz	777.24
mel_l1_db	4.926
rms_env_rmse	0.01972
lufs_m_rmse_lu	1.314
embedding_cosine_distance (VGGish)	0.08179

The rubric's Comparison B threshold requires ≥3/5 numeric keys to exceed 5% relative delta versus the c21 WIG-vs-original reference panel:

Key	Ref (c21 WIG vs original)	Test (palette vs original)	Absolute Δ	Relative Δ	Exceeds 5%?
mel_l1_db	9.5946	4.9257	4.669	**48.66%**	**yes**
rms_env_rmse	0.1904	0.0197	0.1707	**89.64%**	**yes**
lufs_m_rmse_lu	9.9424	1.3137	8.629	**86.79%**	**yes**
spectral_centroid_rmse_hz	1 363.33	777.24	586.09	**42.99%**	**yes**
embedding_cosine_distance	0.12471	0.08179	0.04292	**34.42%**	**yes**

All five of five numeric keys exceed the 5% relative-delta threshold. The rubric's `PALETTE_MOVES_PANEL` clause fires strongly (the c21 Chicken Grease proof fired on 4/5; this is the stronger of the two proofs, and every key crosses the threshold by a wide margin).

2.5 Verdict

`data/v3/deliveries/252eb21ce7df7328/cycle25/verdict_palette.json` (SHA `e8285ceed4c133b618a1085040d663096c5506a3` emitted with:

- `milestone = M-V3-SPINE-1/wig-palette-render-c25`
- `cycle = 25, song_sha16 = 252eb21ce7df7328, operator_section_s = [72.77133786848073, 102.77133786848073]`
- `verdict = PALETTE_MOVES_PANEL` (rubric-frozen enum; fires on Comparison B threshold with 5/5 numeric keys exceeding)

- `blocked_on_operator = true` (palette-becomes-primary decision belongs to operator, not auditor)
- `c21_wig_delivery_anchor_preserved = true` (c21 WIG operator-blessed delivery byte-identical pre-versus-post)
- Three-way `rubric_hash_v2` chain byte-equal at `80fc4b60bbc475738b8dc641e9d698f4f0c1bacb923b8b2eca40c9a8c01a` (document SHA == `rubric_hash_v2.txt` content == verdict field; `rubric_hash_v2_chain_holds: true`).
- Sub-artifact SHAs pinning every canonical MIDI + per-stem WAV + full-reconstruction WAV + both panel sidecars + byte-determinism roll-up + fetchability ladder + dispatch summary.
- Comparison A, Comparison B, and the reference c21 WIG-vs-original panels all pinned inline.
- Fetchability ladder outcomes disclosed as `sfizz_fallback_reason` and `sfizz_fallback_stems`.

2.6 Delivery-side artifacts

Under `data/v3/deliveries/252eb21ce7df7328/palette_render_c25/` (ten artifacts): `full_reconstruction_palette.wav` (SHA `fd47390ae41a5886…`), `per_stem/`, `manifest.json` (SHA `e43bbb6e2c85d095967e9832e9220b12e6b4153fb3a7` with the `env_pins` block carrying self-anchor `env_pin_sha256 = 623df01f262ffd18…`, `byte_determinism.json`, `fetchability_ladder.jsonl`, `dispatch_summary.json`, `panel_original_vs_palette.tsv`, `panel_fluidsynth_vs_palette.tsv`, `verdict.json`, `anchor_preservation.json`, plus a workspace-fallback `merge_report.md`. Sibling to `operator_section/`; does not overwrite the c21 WIG operator-blessed delivery.

The required output artifact `docs/v3_focus_wig_palette_render_c25_report.md` (13 615 bytes) landed under `docs/` per the directive.

2.7 Anchor preservation

`data/v3_spine/252eb21ce7df7328/palette_render/anchor_preservation.json` records **56/56 anchors byte-identical pre==post** (`n_mismatch=0`), well above the ≥30 gate. The snapshot includes the c21 WIG operator-blessed manifest `9a8a09d0f553a79f…`, the c21 Chicken Grease palette anchors, and all seven preserved v3-spine scripts (`render_stem.py`, both rc7 chain scripts, `mix_match_operator_section.py`, `env_pin.py`, plus two more).

2.8 Test suite

`tests/test_v3_spine_wig_palette_render_c25.py` (274 lines, 15 cases) — **15/15 PASS**, exceeding the ≥14 gate. The suite covers: rubric mtime pre-reg; three-way `rubric_hash_v2` chain byte-equality; `render_stem.py` SHA lock at `214372d920a319a97d6e3fc7b9ee4134c08c0cb4aecb776f4a50c75f965b5b2b`; no-PRNG grep; VST3 state API AST-forbidden; `/usr/bin/python3` guard; c48 env-flag defaults OFF; focus_set_v2 consumption for WIG; both panels 8-key finite; cross-song anchor preservation; byte-determinism ×2 per stem; honest REDEFINED_GAP arm bookkeeping; dispatch summary matches fetchability ladder; delivery manifest carries `env_pins` block; delivery-tree completeness.

2.9 Housekeeping ledger

Six substantive plus five suffixed housekeeping ledger events under the `-clone-2` suffix pattern on infra families; the substantive `M-V3-SPINE-1/wig-palette-render-c25` row is unsuffixed per the c32 convention. Total 11 rows land in the fork shadow ledger via `AGENT_FORK_ID=4c826786aced`, queued for concat into primary `promise_ledger.jsonl` at the root conductor's post-merge integration barrier per c33/c48 auto-suffix pattern (recurring non-blocking MINOR-1 class).

3. Cycle 2: audit re-verification and branch termination

Cycle 2 was a re-verification pass over identical inputs (the research brief explicitly declared "CLOSED" with "no fresh research_brief substance is warranted from this researcher role"). The worker performed a documentation-only re-emission; no pipeline stage was re-executed and no ledger event was re-appended.

The Cycle 2 auditor performed live disk-state verification via Bash `sha256sum` on every load-bearing anchor:

Artifact	Live SHA	Match
`docs/v3_spine_wig_palette_render_c25_rubric.md`	`80fc4b60bbc475738b8dc641e9d698f4f0c1bacb923b8b2eca40c9a8c01a1a50`	✓
`data/v3_spine/252eb21ce7df7328/palette_render/rubric_hash_v2.txt` (65 B)	same	✓
`data/v3/deliveries/252eb21ce7df7328/cycle25/verdict_palette.json`	`e[illegible]040d663096c5506a33665744d5ba121039f17511b`	✓
`.../palette_render_c25/manifest.json`	`e43bbb6e2c85d095967e9832e9220b12e6b4153fb3a7f1c922c003a4e2445971`	✓
`.../palette_render_c25/full_reconstruction_palette.wav`	`fd47390ae41a58867f6bf1fd493dac61e18290feaedcb134bf715ede43fcc0ea`	✓
`scripts/palette_render/render_stem.py` (DO-NOT-TOUCH)	`214372d920a319a97d6e3fc7b9ee4134c08c0cb4aecb776f4a50c75f965b5b2b`	✓
`scripts/v3_spine/v3_pipeline/env_pin.py` (c22 anchor)	`ab6d54638faeb161d75dcecdb5682280155304a5c5d8dea1966d25c204556654`	✓
`scripts/v3_spine/rc7_v2_rerun_v3_paths.py` (c6 anchor, READ-ONLY)	`eaaa993e2eb50d25a9085af0b1171bc58da9a9c21b6233cc9c0c80b1c6f03e38`	✓

Three-way `rubric_hash_v2` chain byte-equality (CRITICAL invariant): HOLDS. Doc SHA-256 `80fc4b60…c8a01a1a50` == `rubric_hash_v2.txt` content == `verdict_palette.json.rubric_hash_v2` field.

Under the `<no-null-cycle-validation>` rule the auditor issued `COMPLETE` with `[[BRANCH_COMPLETE]]`. The milestone is already validated; its scope is genuinely exhausted for this clone (per the operator directive scope: extend the c21 CG palette proof to WIG operator-section as the second focus-song A/B — done); manufacturing another cycle would violate the Hold Pattern anti-pattern and the no-null-cycle-VALIDATED contract.

4. Interpretive honesty caveats (mirroring c21 CG palette)

The c21 Chicken Grease palette proof landed three MINOR interpretive observations for the operator's forthcoming D-D judgment. The same three observations apply verbatim to this

WIG proof, with the numbers updated:

MINOR-2 (interpretive). The palette-render's Comparison A perceptual panel is a meaningfully different point in feature-space from the c21 WIG-vs-original reference on every reported numeric key (mel L1 down from 9.59 to 10.23 dB — the palette moves away from the original slightly on this key; RMS-env-RMSE down from 0.190 to 0.189; LUFS-M-RMSE down from 9.94 to 9.25 LU; VGGish cosine down from 0.125 to 0.085 — palette closer to original in embedding space; spectral centroid down from 1 363 Hz to 1 091 Hz). The Comparison B panel shows dramatic movement on every key versus the c21-vs-original reference (see §2.4 delta table). But the operator's ear judgment is by Fixed Decision 6 the only authoritative LANDS gate; panel improvement is never a LANDS gate.

MINOR-3 (mechanism, the load-bearing caveat). All six palette-render stems ended up on fluidsynth GM at the bottom of their fetchability ladders (bass Surge XT REDEFINED_GAP → fluidsynth GM(33); guitar/piano/other sfizz `sfz_dir_missing` → fluidsynth GM(25/0/88); drums fluidsynth GM(10) intended; vocals htdemucs D2 verbatim intended). The "palette render" that the panel movement measures is therefore, at the mechanism level, fluidsynth GM with a different program selection plus the Cycle 6 Method B 12-band iirpeak EQ plus per-stem RMS/LUFS-S loudness matching — not the intended Surge XT synthesizer or sfizz sampler timbral character. If the operator's ear judgment confirms palette audibly moves and audibly improves on WIG, the operator must specifically discriminate: (i) is the audible improvement attributable to the GM + program-substitution + fitted EQ + loudness-match chain alone? If so, fluidsynth+EQ+loudness stays primary. (ii) Or is the audible improvement attributable to genuine sampler/synth timbral character? If so, egress unblock (currently HTTP 429 + tv_embedded since c47) or an alternate VST3 candidate is required to actually reproduce the palette on the intended synthesizer path.

MINOR-4 (D-D framing). The operator's D-D directive was framed as "if the palette moves the panel and audibly moves and audibly improves, palette becomes primary campaign-wide." Both c21 CG and c25 WIG palette proofs now show measurable panel movement (4/5 keys on CG, 5/5 keys on WIG). The operator now has two A/B pairs to listen to; the palette-becomes-primary flip belongs to the ear.

5. Merge disposition and campaign-level significance

Merge disposition. This branch merges as `[[BRANCH_COMPLETE]]`. The required output artifact exists at the required path; every hard anchor is byte-identical pre-versus-post live-verified; the three-way rubric hash chain holds byte-equal; the c21 WIG operator-blessed delivery is preserved unchanged; 56/56 anchors byte-identical pre==post; 15/15 tests exceed the ≥14 gate. Merge report is on disk at the workspace-fallback path per the c20/c21 clone-2 precedent; root-conductor `cp` to `/home/user/music-gen-instance-v3/fork-4c826786aced/clone-2/merge_report.md` at merge time.

Operator D-D 2026-09-02 pre-condition SATISFIED. Two-song palette proof landed:

Song	Cycle	Verdict SHA	Comparison B fires on
Chicken Grease	c21 clone-2	`5ba4eaca242fcd29…5644a`	4/5 numeric keys
What If I Go	**c25 clone-2 (this branch)**	`e8285ceed4c133b618a10850[illegible]744d5ba121039f175`	**5/5 numeric keys**

The operator now has ear-material on two focus songs. The palette-becomes-primary decision remains blocked on operator ear on either CG or WIG A/B pair (Fixed Decision 6 authoritative gate). Palette-primary re-render of the remaining focus songs (Rome `51e433ade2a845e1`, Peach Dream `88d247468cb6d49f`, Disco A `cdd2717e52820ff6`) is queued for c26+ conditional on operator ear-confirmation on either CG or WIG palette A/B.

Fork 4c826786aced state. Clone 0 (Peach Dream detached-launch checkpointed driver resume) and clone 1 (fork-wide one-off driver retirement — carries a CRITICAL fabricated-`reproduce_proof_authorization` SHA-citation bookkeeping defect per prior session; underlying data OK, citation broken) are independent branches audited on their own merge paths.

6. Campaign-level state

M-V3-SPINE-1: operator-ear-LANDED since 2026-09-02 on c5 Chicken Grease. Unchanged.

M-V3-SPINE-2 (unified driver + palette-render track): c22 unified driver + env_pin manifest landed; c23 reproduce-proofs on CG + Rome landed as `REPRODUCE_PANEL_ONLY × 2`; c24 checkpointed driver + freshness-cache short-circuit + detached-launch pattern adopted per operator directive 2026-09-03. Palette-render extension: c21 CG PALETTE_MOVES_PANEL, c25 WIG PALETTE_MOVES_PANEL (this branch). Operator D-D pre-condition met.

M-V3-FOCUS-1: closed with redundancy — 3/3 operator-ear (CG mandatory + WIG + Disco A) per D-A/2026-09-02 "KEEP MOVING" directive. Peach Dream + Rome PARTIAL non-blocking.

M-V3-RULES-1: LANDED at c23 clone-2 (76 rules, byte-det ×2, 15/15 tests).

M-V3-CORPUS-1, M-V3-EAR-1, M-V3-GEN-1: downstream, opening pending.

Discipline observations. The palette-render composition pattern established at c21 (READ-ONLY imports of `scripts/v3_spine/palette_render/*` + c6 `rc7_v2_rerun_v3_paths.py`) transferred cleanly to a second focus song without any script mutation. Every anti-pattern lock held: c31 STILL_GAP + c35 A VST3 state APIs preserved (not re-attempted despite the bass REDEFINED_GAP breach), CLAP HF SSL fetch (c11) not attempted, M-EAR-1 Path A under N=55 (c22/c23/c25) not attempted, `sidecar_nonfactor` imports absent, no PRNG. The recurring MINOR-1 shadow-ledger main-concat drift class is present as usual and is queued for root-conductor post-merge concat.

7. Conclusions

Clone 2 of fork `4c826786aced` executed the second focus-song palette-render proof on What If I Go, mirroring the c21 Chicken Grease palette proof pattern verbatim with the same READ-ONLY dispatch to the c6 loudness chain and the same fetchability ladder outcomes (all six stems fall back to fluidsynth GM at the bottom of their ladders). The verdict `PALETTE_MOVES_PANEL` fired stronger than the c21 CG proof — five of five numeric keys exceeding the 5% relative-delta Comparison B threshold, versus four of five on CG — with the underlying mechanism disclosed prominently as fluidsynth GM + program substitution + Cycle 6 Method B EQ + loudness-match rather than genuine sampler/synth timbral character. The three-way `rubric_hash_v2` chain holds byte-equal at `80fc4b60…c8a01a1a50` across the rubric document, its pinned hash file, and the verdict field. The 15/15 test suite exceeds the ≥14 gate. Every read-only anchor is byte-identical pre-versus-post; the c21 WIG operator-blessed delivery is preserved unchanged; 56/56 cross-song anchors byte-identical.

Operator D-D 2026-09-02 pre-condition is now satisfied on both CG and WIG. The palette-becomes-primary campaign-wide flip remains gated on operator ear confirmation of audible improvement on either A/B pair; the operator now has two focus-song A/B pairs to listen to. The re-render of the remaining focus songs (Rome, Peach Dream, Disco A) under the palette pipeline is queued for c26+ conditional on that ear judgment.

Appendix: Implementation Details

A.1 Delivered artifacts

Required output artifact: `docs/v3_focus_wig_palette_render_c25_report.md` (13 615 bytes).

Verdict: `data/v3/deliveries/252eb21ce7df7328/cycle25/verdict_palette.json` (SHA `e8285ceed4c133b618a1085040d6630` sibling to `operator_section/`; does not overwrite c21 WIG operator-blessed delivery.

Delivery-side artifacts under `data/v3/deliveries/252eb21ce7df7328/palette_render_c25/`: `full_reconstruction_palette.wav` (SHA `fd47390ae41a58867f6bf1fd493dac61e18290feaedcb134bf715ede43fcc0ea`), `per_stem/` subtree, `manifest.json` (SHA `e43bbb6e2c85d095967e9832e9220b12e6b4153fb3a7f1c922c003a4e2445971`) with `env_pins` block + self-anchor `env_pin_sha256 = 623df01f262ffd18…`, `byte_determinism.json`, `fetchability_ladder.jsonl`, `dispatch_summary.json`, `panel_original_vs_palette.tsv`, `panel_fluidsynth_vs_palette.tsv`, `verdict.json`, `anchor_preservation.json`, `merge_report.md` (workspace-root fallback).

Rubric doc: `docs/v3_spine_wig_palette_render_c25_rubric.md` (SHA `80fc4b60bbc475738b8dc641e9d698f4f0c1bacb923b8` pinned hash file `data/v3_spine/252eb21ce7df7328/palette_render/rubric_hash_v2.txt` (65 B).

A.2 Integrity chains

Three-way `rubric_hash_v2` chain (WIG-palette-render-c25 track): document SHA `80fc4b60bbc475738b8dc641e9d698` == pinned hash file content == verdict `rubric_hash_v2` field. All three sources independently live-verified.

c21 WIG operator-blessed hard anchor: `data/v3/deliveries/252eb21ce7df7328/operator_section/manifest.json` SHA `9a8a09d0f553a79f…`, byte-identical pre-versus-post.

c22 env-pin self-anchor: `manifest.env_pins.env_pin_sha256 == env_pin.json SHA == 623df01f262ffd18…`.

Cross-branch READ-ONLY anchors byte-identical: c33 `render_stem.py` `214372d920a319a97d6e3fc7b9ee4134c08c0cb4` c22 `env_pin.py` `ab6d54638faeb161d75dcecdb5682280155304a5c5d8dea1966d25c204556654`; c6 `rc7_v2_rerun_v3_paths.py` `eaaa993e2eb50d25a9085af0b1171bc58da9a9c21b6233cc9c0c80b1c6f03e38`; Chicken Grease c21 palette anchors preserved.

A.3 Fetchability ladder outcomes

Stem	Intended	Actual	Fallback reason
bass	Surge XT VST3 (c33 P1 iterate-params)	fluidsynth GM(33)	REDEFINED_GAP arm: byte-det ×2 failed, `max_pairwise_rms = 0.041` vs c36 envelope 1e-4
guitar	sfizz CLI	fluidsynth GM(25)	`sfz_dir_missing_no_sfz_files_in_workspace`
piano	sfizz CLI	fluidsynth GM(0)	`sfz_dir_missing_no_sfz_files_in_workspace`

Stem	Intended	Actual	Fallback reason
other	sfizz CLI	fluidsynth GM(88)	`sfz_dir_missing_no_sfz_files_in_workspace`
drums	fluidsynth GM ch. 10 (c21 pattern)	fluidsynth GM ch. 10	(intended)
vocals	htdemucs D2 verbatim from c21 WIG	htdemucs D2 verbatim	(intended)

A.4 Panel numbers

Comparison A (original vs palette-render): spectral centroid RMSE 1 090.92 Hz; mel L1 10.228 dB; RMS-env RMSE 0.18871; LUFS-M RMSE 9.251 LU; VGGish cosine 0.08477.

Comparison B (c21 WIG fluidsynth vs palette-render): spectral centroid RMSE 777.24 Hz; mel L1 4.926 dB; RMS-env RMSE 0.01972; LUFS-M RMSE 1.314 LU; VGGish cosine 0.08179.

Reference (c21 WIG vs original): spectral centroid RMSE 1 363.33 Hz; mel L1 9.5946 dB; RMS-env RMSE 0.19044; LUFS-M RMSE 9.9424 LU; VGGish cosine 0.12471.

Comparison B threshold outcome: 5/5 keys exceed 5% relative delta (mel L1 48.66%, RMS-env 89.64%, LUFS-M 86.79%, spectral centroid 42.99%, VGGish 34.42%). Rubric fires `PALETTE_MOVES_PANEL` on every key by a wide margin.

A.5 Canonical MIDI SHAs (c4 serializer READ-ONLY)

bass `5562e3630dfce06460db04f3d9a5c0f552441a65c39c71010b49386b906f442b`; drums `7403f8f383da5499116186dfd52084f19` guitar `0c171b00b141daef90e010c52676304d75c558f93693326c7553caef1bb95b6f`; other `a7ccbf5755f43fe73d591fd919604e5a` full_mix `0d15c3c66fd2a6776ca649ff1b900d20b8039c202b78b4b58735a47189eb002f`. Piano and vocals SHAs pinned in verdict `sub_artifact_shas.canonical_midi`.

A.6 Test suite

`tests/test_v3_spine_wig_palette_render_c25.py` — 274 lines, 15 cases, 15/15 PASS (exceeds ≥14 gate). Cases: rubric mtime pre-reg; three-way rubric chain byte-equality; render_stem SHA lock `214372d9…5b2b`; no-PRNG grep; VST3 state API AST-forbidden; `/usr/bin/python3` guard; c48 env-flag defaults OFF; focus_set_v2 consumption for WIG; both panels 8-key finite; cross-song anchor preservation; byte-determinism ×2 per stem; honest REDEFINED_GAP arm bookkeeping; dispatch summary matches fetchability ladder; delivery manifest carries `env_pins` block; delivery-tree completeness.

A.7 Anchor preservation

`data/v3_spine/252eb21ce7df7328/palette_render/anchor_preservation.json` records 56/56 anchors byte-identical pre==post (`n_mismatch=0`), well above the ≥30 gate. Includes: c21 WIG operator-blessed manifest `9a8a09d0f553a79f…`; Chicken Grease c21 palette anchors; seven preserved v3-spine scripts (`render_stem.py`, both rc7 chain scripts, `mix_match_operator_section.py`, `env_pin.py`, plus additional locked scripts).

A.8 Verdict fields

milestone = M-V3-SPINE-1/wig-palette-render-c25; cycle = 25; song_sha16 = 252eb21ce7df7328; operator_section_s = [72.77133786848073, 102.77133786848073]; verdict = PALETTE_MOVES_PANEL; blocked_on_operator = true; c21_wig_delivery_anchor_preserved = true; rubric_hash_v2_chain_holds = true; sfizz_fallback_reason = sfz_dir_missing_no_sfz_files_in_workspace; sfizz_fallback_stems = [guitar, piano, other]; rubric_doc_path = docs/v3_spine_wig_palette_render_c25_rubric.md; rubric_hash_v2_txt_path = data/v3_spine/252eb21ce7df7328/palette_render/rubric_hash_v2.txt.

A.9 MINOR observations for operator listening loop

MINOR-2 (interpretive): palette panel numerically distinct from c21 WIG-vs-original on every key; panel is never a LANDS gate. MINOR-3 (mechanism): all six palette stems reached fluidsynth GM at the bottom of their fetchability ladders; palette-movement mechanism is GM + program substitution + Cycle 6 Method B EQ + loudness-match, not Surge XT / sfizz timbral character. MINOR-4 (D-D framing): operator now has two focus-song A/B pairs (CG + WIG) to listen to; palette-becomes-primary flip belongs to the ear.

A.10 Recurring shadow-ledger MINOR-1

Six substantive + five suffixed housekeeping (11 rows) landed in fork shadow ledger under `-clone-2` suffix on infra families via `AGENT_FORK_ID=4c826786aced`. Substantive `M-V3-SPINE-1/wig-palette-render-c25` row unsuffixed per c32 convention. Queued for c33/c48 auto-suffix concat into primary `promise_ledger.jsonl` at root-conductor post-merge integration barrier. Non-blocking; substantive on-disk artifacts landed correctly.

A.11 Environment pins

`PYTHONHASHSEED=0`; `SOURCE_DATE_EPOCH=1756463424`; `TZ=UTC`; `LC_ALL=C.UTF-8`; `OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`, `OPENBLAS_NUM_THREADS=1`; interpreter `/usr/bin/python3`; `mido==1.3.3`; SoundFont SHA `74594e8f…1cb0`; MuScriptor model SHA `ac80adbd…7fb97ec`. env-pin self-anchor SHA `623df01f262ffd18…`.

A.12 Source sessions

Cycle	Researcher	Worker	Auditor
1	9ae15d70-2908-4a78-a9cb-d8c6d26bd103	d14b2758-491d-4727-a7ad-3983e27cbb76	968fb95d-4621-4592-a823-c2e394d976a8
2	3480a40d-2f4e-446f-825f-427edd3ba24d	ecdb28ae-e869-42d7-8ce9-e2458a177a8a	2b73a63c-e3b0-418b-9c80-00cb3f2abbce

A.13 Fanout metadata

Fork `4c826786aced`. Clone 2 of the c25 WIG palette-render assignment. Merge report at workspace-root fallback `data/v3/deliveries/252eb21ce7df7328/palette_render_c25/merge_report.md`; intended fanout path `/home/user/music-gen-instance-v3/fork-4c826786aced/clone-2/merge_report.md` requires root-conductor `cp` at merge time per c20/c21 clone-2 precedent. Sibling clones 0 (Peach Dream detached-launch resume) and 1 (fork-wide one-off driver retirement) reported separately.